

PYTHON PROGRAMMING ASSIGNMENT

Build Your Own Rule-Based Python Chatbot

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1. Aim

To build a simple rule-based conversational chatbot in Python that interacts continuously with the user, recognizes different messages, provides appropriate responses, and allows the user to exit the conversation.

2. Requirements Covered

1. Ask the user's name before the main conversation.
2. Use while True: to keep the conversation running.
3. Use input() to receive messages.
4. Clean the input using .lower().strip().
5. Recognize at least two greetings, such as hello and hi.
6. Answer at least three different questions or topics.
7. Use if / elif / else for decision making.
8. Use break to exit when the user types bye or exit.
9. Use the user's name in at least one response.
10. Provide a useful response for unknown messages using else.
11. Use the in operator so variations such as hello bot trigger a greeting.

3. Algorithm / Logic

1. Start the chatbot and ask the user's name.
2. Display a welcome message.
3. Use while True to continuously receive messages.
4. Use input().lower().strip() to clean each message.
5. Check bye or exit and use break to terminate.
6. Check greetings using the in operator.
7. Check supported questions/topics and provide suitable responses.
8. Use else for unknown messages and repeat until exit.

4. Python Source Code

```
# Simple Rule-Based Python Chatbot

print("=====")
print("          WELCOME TO MY PYTHON CHATBOT          ")
print("=====")

name = input("What is your name? ").strip()

print(f"Nice to meet you, {name}!")
print("You can say hello, ask questions, or type 'bye' or 'exit' to stop.")
print("-" * 45)

while True:
    user_input = input(f"{name}: ").lower().strip()

    if user_input == "bye" or user_input == "exit":
        print(f"Chatbot: Goodbye, {name}! Have a great day!")
        break

    elif "hello" in user_input or "hi" in user_input:
        print(f"Chatbot: Hello {name}! How are you?")

    elif "your name" in user_input or "who are you" in user_input:
        print("Chatbot: I am a simple rule-based Python chatbot.")

    elif "python" in user_input:
        print("Chatbot: Python is a popular programming language that is easy to learn.")

    elif "help" in user_input:
        print("Chatbot: I can respond to greetings, answer Python questions, and help with simple topics.")

    elif "how are you" in user_input:
        print("Chatbot: I am doing great! Thanks for asking.")

    else:
        print("Chatbot: Sorry, I don't understand that yet.")
        print("Chatbot: Try saying hello, asking about Python, typing help, or saying bye.")
```

5. Sample Execution / Demonstration

The supplied Google Colab output screenshot is included below as evidence of the chatbot execution.



```
=====
          WELCOME TO MY PYTHON CHATBOT
          =====
What is your name? Raghava

Nice to meet you, Raghava!
You can say hello, ask questions, or type 'bye' or 'exit' to stop.
-----
Raghava: hello
Chatbot: Hello Raghava! How are you?
Raghava: hello bot
Chatbot: Hello Raghava! How are you?
Raghava: what is python
Chatbot: Python is a popular programming language that is easy to learn.
Raghava: bye
Chatbot: Goodbye, Raghava! Have a great day!
```

Figure 1. Google Colab output of the Python chatbot.

6. Test Cases

Test Input	Expected Result
hello	Greeting response
HELLO	Greeting response after lower()
hi	Greeting response
hello bot	Greeting response using in operator
what is python	Python-related response
who are you	Chatbot identity response
unknown message	Useful response from else
bye / exit	Goodbye message and program terminates

7. Conclusion

The chatbot successfully follows the INPUT -> PROCESS -> RESPOND -> REPEAT -> EXIT flow. It continuously accepts user messages, handles greetings and multiple topics, uses conditional logic, supports flexible matching, responds to unknown inputs, and exits with bye or exit.

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